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Examining the Effect of Bat Shape on the Exit Velocity of a Batted Ball

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ABSTRACT

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1. INTRODUCTION

The most crucial moment in baseball occurs within roughly 5 milliseconds. In that time, a batter swings at a pitch and, if they're lucky, makes contact with the ball. The ball flies off the bat with a certain velocity and spin, which will determine if those ≈5 milliseconds will result in a hit, an out (or two), a foul, or even a home run. Physically, these 5 milliseconds contain complex momentum and energy transfers, which depends on the incoming ball, bat swing profile, and the properties of the bat itself. While it's been long known to the baseball world that different bat materials can produce drastically different results, the introduction of the Torpedo bat in 2025 proved that bat shape is also a variable that should be optimized.

The Torpedo bat, designed by Aaron Leanhardt and pioneered with the New York Yankees, refocuses mass towards the "sweet spot" of the bat. For players with a high sweet spot rate, the increase in mass should cause their batted balls to have higher exit velocities and travel further. In theoretical calculations presented on the baseball analysis site Fangraphs, Prof. Alan Nathan predicts that a Torpedo bat should have a wider sweet spot and higher exit velocities farther from the tip compared to a standard bat (A. Nathan 2025). However, physical tests of the Torpedo bat have yet to reproduce Nathan's findings. The Torpedo bat is still so novel that the desired data on exit velocities for batted balls is not yet robust for testing Nathan's theory.

Nathan's calculations are based on the bat-ball collision model he presented in A. M. Nathan (2000), which elaborates on the work done by L. L. Van Zandt (1992). Van Zandt proposes a baseball bat is not a rigid body. Rather, it oscillates or "wiggles" in response to a collision. He models the bat as an accumulation of many small slices (sliced lengthwise, so each cross-section is circular), which are coupled to the adjacent slices. As a result, a small displacement in one bat slice propagates along the bat. The bat also shears, so each slice can move angularly with respect to the axis of the bat. The frequencies of the propagation depend not only on the stiffness and density of the bat, but also its moment of inertia tensor and each slice's cross-sectional area. Van Zandt solves for the normal modes of the bat and demonstrates how the bat's shape deforms in time as a response to a collision.

Nathan elaborates on Van Zandt's work by proposing a different model for the ball and subsequent collision. Nathan models the ball as a nonlinear lossy spring, such that a collision of the ball with a rigid body causes the ball to deform and expand. The non-elastic deformation causes much of the initial kinetic energy of the ball to be lost. Nathan analyzes the energy transfers involved in the collision and ensures that, overall, energy is conserved. Constrastingly, Van Zandt does not account for energy conservation in his model. Nathan argues that energy conservation explains why his model better agrees with observational data compared to Van Zandt.

Since Nathan's paper in 2000, computational methods in physics have improved drastically. The capability of modern computers to integrate complex systems and simulate theoretical results motivate a second look at Nathan's equations. In this paper, we re-integrate the differential equations proposed in A. M. Nathan (2000) to create a bat-ball model tool that can be broadly applied to different bat shapes. We compare our results to Nathan's findings for a standard bat and extend his model to the Torpedo bat. In this study, we determine if our integration of the Torpedo bat agrees with Nathans analysis in Fangraphs, or if we can explain discrepancies between his model and observed Torpedo-batted balls.

In this paper...

2. BAT-BALL COLLISION MODEL

2.1. *The Bat*

In accordance with Nathan and Van Zandt, we model the baseball bat discretely. We align the main axis of symmetry of the bat with the z-axis, and slice the bat into N sections of thickness dz perpendicular to the axis (see Figure 1). Each slice of the bat has two degrees of freedom. A slice can displace vertically from the equilibrium position and also angularly with respect to the vertical. We establish coordinates such that ith slice has vertical displacement y_i and angular displacement ϕ_i (see Figure **??**). In our examination, we solve for $\Phi_i = \phi_i dz$ to maintain units of length. In total, our bat has $2N$ degrees of freedom.

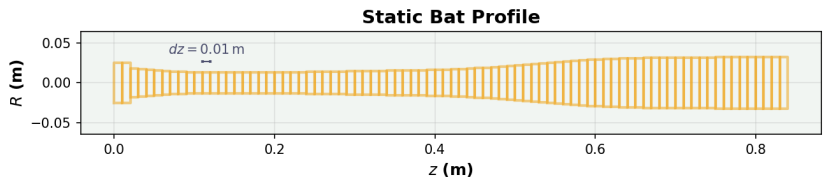

Figure 1. An example radial profile of a static baseball bat. The bat is split lengthwise into N slices of width dz. For this example, we take $N = 84$ and $dz = 0.1$cm.

Crucially, each bat has a measured radial profile $\{R_i\}$, where R_i represents the radius of the ith slice of the bat. The batter's hands grasp the nob, or the 0th slice of the bat. The Nth slice is refered to as the tip. For a solid wooden bat, each slice then has cross-sectional area $A_i = \pi R_i^2$ and moment of intertia $I_i = \pi R_i^4/4$. Our examination also requires three observed parameters, Young's modulus Y_i, shear modulus S_i, and density ρ_i. A bat is typically constructed from a continuous wooden piece, so we take the reasonable approximation that these parameters are uniform. Henceforth, the subscripts are dropped, and we refer to these parameters as Y, S, ρ.

Van Zandt derives the equations of motion for the bat from physics first principles. If a single slice is displaced, the neighbouring slices will impose a restoring force to equilibrium. He relates this restoring force to the torques throughout the bat and Newton's second law. The results align with Timoshenko's beam equations for oscillations of a free beams. He obtains $2N$ coupled 2nd order ordinary differential equations. These are:

$$\ddot{y}_i = \frac{S}{\rho(dz)^2}\left[y_{i+1} + y_{i-1} - 2y_i + \left(\frac{A_i - A_{i-1}}{A_i}\right)(y_i - y_{i-1} + \Phi_{i-1}) + (\Phi_i - \Phi_{i-1})\right] \tag{1}$$

and

$$\begin{aligned}\ddot{\Phi}_i =& \frac{Y}{\rho(dz)^2}\left[(\Phi_{i-1} + \Phi_{i+1} - 2\Phi_i) + \left(\frac{I_i - I_{i-1}}{I_i}\right)(\Phi_i - \Phi_{i-1})\right] \\ &- \left(\frac{S}{2\rho I_i}\right)[A_i(y_{i+1} - y_{i-1} + \Phi_i + \Phi_{i-1}) - (A_i - A_{i-1})(y_i - y_{i-1} + \Phi_{i-1})].\end{aligned} \tag{2}$$

In baseball, the knob of the bat is held in the batter's hands. However, the propogation time of a wave along the bat is much slower than the collision time with the ball. On time scales relevant to our problem, we conclude it is appropriate to consider the bat free on both ends. The first and last slices respond to no external force or torque.

The equations of motion are linear and contain no first order terms, so we can rewrite them as:

$$\ddot{\vec{\psi}} = \mathbf{H}\vec{\psi} \tag{3}$$

Where $\mathbf{H}$ is a $2N \times 2N$ non-symmetric matrix, and

$$\vec{\psi} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \\ \Phi_1 \\ \vdots \\ \Phi_N \end{bmatrix}. \tag{4}$$

We can imagine $\mathbf{H}$ as a concatenation of four quadrants, each describing how a $\ddot{\psi}_i$ component depends on a first-order term. The first quadrant relates $\ddot{y}_i$ to y_k, the second relates $\ddot{y}_i$ to Φ_k, the third relates $\ddot{\Phi}_i$ to y_k, and the fourth relates $\ddot{\Phi}_i$ to Φ_k (See Figure 2). Since each slice is only coupled to its neighbours, we expect each quadrant of $\mathbf{H}$ to be tridiagonal. First, we define

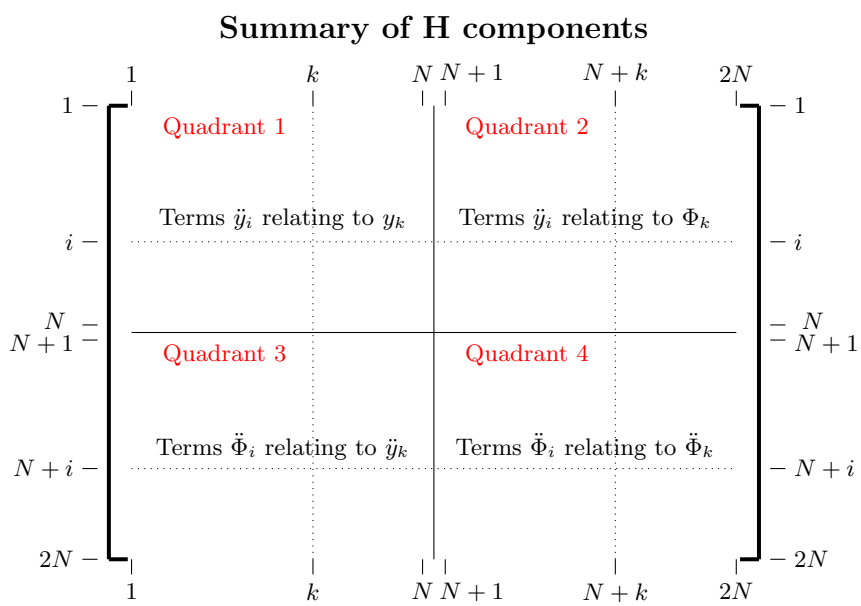

Figure 2. Schematic explaining the formation of the matrix $\mathbf{H}$, which is the defining matrix of our equations of motion. $\mathbf{H}$ contains four quadrants, each with a coefficient that relates an element of $\ddot{\psi}$ to ψ. The first quadrant relates $\ddot{y}_i$ to y_k, the second relates $\ddot{y}_i$ to Φ_k, the third relates $\ddot{\Phi}_i$ to y_k, and the fourth relates $\ddot{\Phi}_i$ to Φ_k. Each quadrant has shape $N \times N$, yielding $\mathbf{H}$ to have shape $2N \times 2N$.

$$\Lambda \equiv \frac{S}{\rho(dz)^2} \qquad\qquad \Upsilon \equiv \frac{Y}{\rho(dz)^2}. \tag{5}$$

We first consider the rows $i \in \{2, 3, ..., N-1\}$, the central rows of $\mathbf{H}$, which are independent of the boundary conditions imposed on the bat. We also let $j = i + N$ as an index for quadrants 2, 3, and 4. We obtain components of $\mathbf{H}$ from Equations 1 and 2, as outlined in Table 1.

Next, we consider the edges of each quadrant. Invoking our boundary conditions, we obtain the components outlined in Table

All other components of $\mathbf{H}$ are uniformly zero.

	Quadrant 1	**Quadrant 2**
Diagonal	relevant indices $\{(i,i)\}$	relevant indices $\{(i,j)\}$
Upper	$H_{i,i+1} = \Lambda$	$H_{i,j+1} = \frac{\Lambda}{2}$
Main	$H_{i,i} = -\Lambda\left(1 + \frac{A_{i-1}}{A_i}\right)$	$H_{i,j} = \frac{\Lambda}{2}\left(1 - \frac{A_{i-1}}{A_i}\right)$
Lower	$H_{i,i-1} = \Lambda\frac{A_{i-1}}{A_i}$	$H_{i,j-1} = -\Lambda\frac{A_{i-1}}{2A_i}$
	Quadrant 3	**Quadrant 4**
Diagonal	relevant indices $\{(j,i)\}$	relevant indices $\{(j,j)\}$
Upper	$H_{j,i+1} = -\Lambda\frac{A_i(dz)^2}{2I_i}$	$H_{j,j+1} = \Upsilon - \Lambda\frac{A_i(dz^2)}{4I_i}$
Main	$H_{j,i} = \Lambda\frac{A_i(dz)^2}{2I_i}\left(1 - \frac{A_{i-1}}{A_i}\right)$	$H_{j,j} = -\Upsilon\left(1 + \frac{I_{i-1}}{I_i}\right) - \Lambda\frac{A_i(dz)^2}{4I_i}\left(1 + \frac{A_{i-1}}{A_i}\right)$
Lower	$H_{j,i-1} = \Lambda\frac{A_{i-1}(dz)^2}{2I_i}$	$H_{i,j-1} = -\Upsilon\left(\frac{I_{i-1}}{I_i}\right) - \Lambda\frac{A_i(dz)^2}{4I_i}$

Table 1. The central components of the marix **H**.

	Quadrant 1	**Quadrant 2**
Boundary	relevant indices $\{(i,i)\}$	relevant indices $\{(i,j)\}$
Knob	$H_{1,1} = -\Lambda$	$H_{1,N+1} = \frac{\Lambda}{2}$
	$H_{1,2} = \Lambda$	$H_{1,N+2} = \frac{\Lambda}{2}$
Tip	$H_{N,N-1} = \Lambda\frac{A_{N-1}}{A_N}$	$H_{N,2N-1} = -\Lambda\frac{A_{N-1}}{2A_N}$
	$H_{N,N} = -\Lambda\frac{A_{N-1}}{A_N}$	$H_{N,2N} = -\Lambda\frac{A_{N-1}}{2A_N}$
	Quadrant 3	**Quadrant 4**
Boundary	relevant indices $\{(j,i)\}$	relevant indices $\{(j,j)\}$
Knob	$H_{N+1,1} = -\Lambda\frac{A_N(dz)^2}{4I_N}$	$H_{j,j+1} = \Upsilon - \Lambda\frac{A_i(dz^2)}{4I_i}$
Main	$H_{j,i} = \Lambda\frac{A_i(dz)^2}{2I_i}\left(1 - \frac{A_{i-1}}{A_i}\right)$	$H_{j,j} = -\Upsilon\left(1 + \frac{I_{i-1}}{I_i}\right) - \Lambda\frac{A_i(dz)^2}{4I_i}\left(1 + \frac{A_{i-1}}{A_i}\right)$

Table 2. The boundary components of the marix **H**.

2.2. *System Integration*

We begin by considering only the bat, without a model for the ball. We take a small perturbation of the kth slice and examine the response wave as it propogates along the bat. To integrate, we employ a Runge-Kutta 4th order integration technique across a uniform time array for a long total time compared to typical bat-ball collision times.

Fourier decomposition of the results will illustrate the normal frequencies of the bat.

3. RESULTS

4. DISCUSSION

ACKNOWLEDGMENTS

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